Fiscal Year:	FY 2014	**Task Last Updated:**	FY 03/05/2014
PI Name:	Schiestl, Robert H Ph.D.		
Project Title:	Effects of Intestinal Microflora on High-LET Radiation Mediated Toxicity and Genomic Instability		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8347	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	11/01/2010	**End Date:**	10/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Borneman, James (University of California, Riverside) Berry, David (University of Vienna)		
Grant/Contract No.:	NNX11AB44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our proposal addresses effects of radiation quality on carcinogenic processes, processes of degenerative diseases, and individual radiation sensitivity. Specifically, we propose to investigate effects of high linear energy transfer (LET)-radiation (space radiation), using high energy protons, silicon ions, and iron ions on immediate and delayed DNA damage, inflammation, and oxidative stress as well as cancer in mice with different compositions of intestinal microbiota. Inflammation and oxidative stress are both causative and associated with several degenerative diseases. Since we will monitor inflammation as well as immunologically regulated anti-inflammatory responses and oxidative stress in mice, we may be able to provide insight into mechanisms of the etiology of radiation-induced degenerative diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Almost nothing is currently known about the effect of the intestinal microbiota on radiation sensitivity, DNA damage, and cancer. Since people have different compositions of intestinal microbiota, our proposal will also contribute to the understanding of individual susceptibility to radiation. If we found a significant effect of intestinal microbiota on radiosensitivity, it would have a major impact on risk assessment. Finally, changing the intestinal microbiota is a non-invasive and easy treatment that could result in a major benefit to astronauts if we identify specific bacteria which alleviate or enhance radiation damage. A simple dose of a mix of probiotics to their diet may counteract a possible loss of intestinal microbiota during extensive space flights.
Task Progress:	Ionizing space radiation causes oxidative DNA damage and triggers oxidative stress responses, and compromised DNA repair mechanisms can lead to increased risk of carcinogenesis. Young adult mice with developed innate and adaptive immune systems that harbour either a conventional intestinal microbiota (CM) or an intestinal microbiota with a restricted bacterial composition (RM) were irradiated with a total dose of 1 Gy delivered by high energy protons (2.5 GeV/n, LET=2 keV/µm), or silicon or iron ions (850 MeV/n, LET = about 50 keV/µm and 1 GeV/n, LET=150 keV/µm, respectively). Microbial compositions were classified by pyrosequencing bacterial taxa of feces. Six hours after whole-body irradiation acute chromosomal DNA lesions were observed for RM mice but not CM mice. In RM mice, four weeks after a single exposure to radiation a decrease of the antioxidant glutathione was measured in whole blood. Elevated levels of phosphorylated gamma-H2AX were observed in irradiated compared to non-irradiated RM mice (and both, irradiated and non-irradiated CM mice had intermediate levels of gamma-H2AX phosphorylation). High throughput sequence analysis of rRNA genes identified mucosa-associated bacterial phylotypes that were colony-specific for RM and CM in ataxia telangiectasia-mutated (ATM)- deficient mice. We conclude that the variable genotoxic endpoints correlate with intestinal microbiota composition and we identified two indicator phylotypes, Lactobacillus johnsonii and Porphyromonas asaccharolytica, as being different in abundance between RM and CM. Conventional microbiota is associated with low levels of radiation-induced DNA damage but is also required for a cyto-protective cellular response against oxidative DNA damage.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Maier I, Berry DM, Borneman J, Schiestl RH. "Molecular oxidative stress and systemic genotoxicity studied in conventional pathobiont-rich and restricted intestinal microbiota mice." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Maier I, Berry DM, Borneman J, Schiestl RH. "Conventional Pathobiont-rich intestinal microbiota protect against molecular oxidative stress and systemic genotoxicity induced by high energy protons." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Articles in Peer-reviewed Journals	Maier I, Berry DM, Schiestl RH. "Intestinal microbiota reduces genotoxic endpoints induced by high-energy protons." Radiat Res. 2014 Jan;181(1):45-53. http://dx.doi.org/10.1667/RR13352.1 ; PubMed PMID: 24397477 , Jan-2014
Articles in Peer-reviewed Journals	Maier I, Schiestl RH. "Evidence from animal models: Is a restricted or conventional intestinal microbiota composition predisposing to risk for high-LET radiation injury?" Radiat Res. 2015 Jun;183(6):589-93. https://doi.org/10.1667/RR13837.1 ; PubMed PMID: 26010710 , Jun-2015
Articles in Peer-reviewed Journals	Maier I, Liu J, Ruegger PM, Deutschmann J, Patsch JM, Helbich TH, Borneman J, Schiestl RH. "Intestinal bacterial indicator phylotypes associate with impaired DNA double-stranded break sensors but augmented skeletal bone micro-structure." Carcinogenesis. 2020 Jun 17;41(4):483-9. https://doi.org/10.1093/carcin/bgz204 ; PMID: 31840161 , Jun-2020
Articles in Peer-reviewed Journals	Yamamoto ML, Maier I, Dang AT, Berry D, Liu J, Ruegger PM, Yang JI, Soto PA, Presley LL, Reliene R, Westbrook AM, Wei B, Loy A, Chang C, Braun J, Borneman J, Schiestl RH. "Intestinal bacteria modify lymphoma incidence and latency by affecting systemic inflammatory state, oxidative stress, and leukocyte genotoxicity." Cancer Res. 2013 Jul 15;73(14):4222-32. PMID: 23860718 , Jul-2013